MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)
Kokapet (Village), Gandipet (Mandal), Hyderabad -500 075.

MR-22

10

Academic Year: 2022-23
B.Tech. - SEMESTER - II
MID - TERM EXAMINATION - I
PART A: OBJECTIVE EXAMINATION

Course : **Environmental Science** Time : **30 Min**
Date & Session : **17/06/2023, FN** Maximum Marks : **10**
Branch & Section: **CE,ME,MCT,MME,EEE&ECE** Roll No:

Note: Answer All Questions. All Questions Carry Equal Marks.

Q.No.	Stem of the Question	A	CO
1	Which of the following bacteria is not associated with Nitrogen cycle (a) Rhizobium b) E. Coli c) Azotobacter d) Nitrobacter		1
2	A network of food chains which are interconnected in an Ecosystem is called A) Food chain B) Ecological pyramid C) Food energy D) Food web		1
3	Primary Consumers in an Ecosystem are also called A) Carnivores B) Omnivores C) Decomposers D) Herbivores		1
4	Population tends to get stabilized when the system reaches a) Primary production b) Homeostasis c) Carrying capacity d) All of the above		1
5	Which of the following is responsible for desertification? (a) Deforestation (b) Overgrazing (c) Mining (d) All of these		2
6	Identify the Non-renewable Source of Energy from the Following A) Hydro power B) Solar Energy C) Wind power D) Coal		2
7	Biomass is (a) Conventional energy (b) Non-Conventional energy (c) Both (d) None of the above		2
8	The energy produced by the hydropower plant is_____ (a) Non-polluting and non-renewable (b) Polluting and non-renewable (c) Non-polluting and renewable (d) Polluting and renewable		2
9	Biodiversity means (a) Various Plants (b) Diversity in living Organisms (c) Ecosystem (d) Ecology		3
10	The concept of Biodiversity was introduced by (a) Hearnst Haeckel (b) Walter G. Rosen (c) Edward O. Wilson (d) Madhav Gadgil		3

		A	CO
11	Increasing concentrations of pollutants at each trophic level is called _____		1
12	Movement of nutrients in an ecosystem is cyclic while flow of energy is _____		1
13	Biotic and abiotic components of an ecosystem influence each other and are linked through energy flow and matter cycling. **(True/False)**		1
14	Conversion of Ammonia to nitrate is known as_____		1
15	_____and _____are examples of Non- renewable energy resources.		2
16	Match the following:	( )	2

List-I	List-II
(i) Sunlight	(A) Non-renewable
(ii) Coal	(B) Renewable
(iii) Heavy rainfall	(C) Drought
(iv) Overgrazing	(D) Flood

The correct answer is

	(i)	(ii)	(iii)	(iv)
(a)	(A)	(B)	(C)	(D)
(b)	(A)	(C)	(B)	(D)
(c)	(B)	(A)	(D)	(C)
(d)	(C)	(B)	(A)	(D)

17 Networking of rivers is being proposed at national level to deal with the problem of floods. **(True/False)** 2

18 Match the following: ( ) 2

List-I	List-II
(i) Sariska tiger reserve	(A) Forest resources
(ii) Narmada valley project	(B) Mineral resources
(iii) Tehri dam	(C) Water resources
(iv) Chernobyl	(D) Energy resources

The correct answer is

	(i)	(ii)	(iii)	(iv)
(a)	(A)	(B)	(C)	(D)
(b)	(B)	(C)	(A)	(D)
(c)	(B)	(A)	(D)	(C)
(d)	(C)	(B)	(A)	(D)

19 5th June is observed as _____ 3

20 The river Ganga is considered as _____value of biodiversity. 3

A - Answer **CO** – Course Outcome

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)
Kokapet (Village), Gandipet (Mandal), Hyderabad 500 075.

B.Tech. –SEMESTER - II **MR-22**

MID-TERM EXAMINATION - I

PART B: DESCRIPTIVE EXAMINATION

Course : **Environmental Science** Time: **90 Min**

Date & Session : **17/06/2023, FN** Maximum Marks: **20**

Branch & Section: **CE, ME, MCT, MME, EEE & ECE** RollNo:

Note: Answer any FOUR full questions. All questions carry equal marks. **(4X5M=20Marks)**

Q.No.	Stem of the Question	M	L	CO
1.a)	Enumerate the Classification of ecosystem	3	2	1
b)	Summarize socio-economical impacts of Floods and Droughts	2	4	2
2.a)	Define Biogeochemical cycle? Explain Carbon cycle in detail.	3	1	1
b)	List out Renewable energy sources. Explicit their advantages.	2	3	2
3.a)	Define food chain, food web? Give with suitable examples	3	4	1
b)	Discuss consequences of over exploitation of mining?	2	1	2
4.a)	Mention the benefits and problems associated with the construction of Big Dams.	3	2	2
b)	Explain Ecosystem values and services.	2	3	1
5.a)	Define Genetic, Species and Ecosystem biodiversity with examples.	3	3	3
b)	What are Ecological Pyramids? Discuss about the Pyramid of energy.	2	2	1
6.a)	What do you mean by consumptive use value, productive use value and Ethical value of biodiversity?	3	2	3
b)	Summarize the benefits of forests and write the ill effects of deforestation	2	1	2

M:Marks; L:Bloom'sTaxonomyLevel; CO:CourseOutcome

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)
Kokapet (Village), Gandipet (Mandal), Hyderabad 500 075.

MR-22

B.Tech. – SEMESTER - I
MID-TERM EXAMINATION - I
PART B: DESCRIPTIVE EXAMINATION

Course : **Environmental Science** Time : **90 Min**
Date & Session : **18/11/2023, FN** Maximum Marks : **20**
Branch & Section: **CSE, IT, CSB, CSD, CSM.** Roll No:

Note: Answer any FOUR full questions. All questions carry equal marks. **(4X5M=20 Marks)**

Q.No.	Stem of the Question	M	L	CO	PO
1.a)	Describe the structure of an Ecosystem.	3	1	1	1
b)	List the benefits and problems associated with big dams.	2	2	2	2
2.a)	Define the concept of biogeochemical cycle. Elaborate the processes of Nitrogen cycle with a diagram.	3	1	1	1
b)	Analyze the reasons behind deforestation and its consequences on the environment.	2	3	2	7
3.a)	Explain Food chain & Food Web with suitable examples. Give the significances.	3	2	1	1
b)	What is flood? Analyze the socio-economic implications arising from the flood conditions.	2	1	2	1
4.a)	Enumerate the advantages of renewable energy resources in terms of its environmental impacts and sustainability.	3	2	2	7
b)	Illustrate the Universal energy flow model of Ecosystem	2	2	1	1
5.a)	Explain Genetic, Species and Ecosystem biodiversity with examples	3	1	3	1
b)	Evaluate environmental effects of extraction of Mineral Resources	2	2	1	2
6.a)	Discuss about Consumptive and Social values of biodiversity with examples for each.	3	1	3	1
b)	What is drought? Analyze the socio-economic implications arising from the prolonged drought conditions.	2	4	2	3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)
Kokapet (Village), Gandipet (Mandal), Hyderabad - 500 075.

MR-22
10

Academic Year: 2022-23
B.Tech./M.Tech. – SEMESTER: I
MID - TERM EXAMINATION: I
PART A: OBJECTIVE EXAMINATION

Course : **Environmental Science** Time : **30 Min**

Date & Session : **18.11.2023 (FN)** Maximum Marks : **10**

Branch & Section: **CSE-1,2,3, CSB, IT, CSM and CSD** Roll No:_____

Note: Answer All Questions. All Questions Carry Equal Marks.

Q.No.	Stem of the Question	A	CO
1.	The organisms which feed on dead matter and Convert into nutrients are are called _____ a) Producers b) Detrivores c) Carnivores d) Decomposers		1
2.	The accumulation of non-biodegradable matter or chemicals in living organisms is known as a) Ecological balance b) Bioaccumulation c) Carrying capacity d) Bio-magnification		1
3.	The amount of radiant energy stored in producers level by photosynthesis is called_____ a) Primary production b) Homeostasis c) Carrying capacity d) All of the above		1
4.	Total number of living organisms at the various trophic levels of a food chain is depicted by a) pyramid by Numbers b) pyramid by Energy c) pyramid by Mass d) All of these		1
5.	Which of the following is unidirectional in an ecosystem. a) Flow of energy b) Flow of nutrients c) Flow of matter d) All of these		2
6.	Which of the following are nonrenewable energy sources _____ a) Coal b) Petroleum c) Natural Gas d) All of these		2
7.	Solar energy can be harnessed by using the principle of _____ a) Photoelectric effect b) Photo Oxidation c) Photo Reduction d) None of these		2
8.	Identify the environmental damage due to mining _____ a) Devegetation b) Ground subsidence c) Air and water pollution d) All of these		2
9.	Different varieties of species within the same species is called _____ biodiversity. a) Genetic b) Species c) Ecosystem d) Both b&c		3
10.	The direct use value of biodiversity is/are _____ a) Food b) Fuels c) Medicines d) All of these		3
11.	The organisms which directly feed on producers are called _____		1
12.	The process of eating and being eaten in an ecosystem is called _____		1
13.	Wind energy can be harnessed by using _____		1
14.	Nitrogen fixing bacteria present in root nodules of legume plants. _____ (True / False)		1
15.	Gross Primary Productivity (GPP) = Net Primary Productivity (NPP) + Respiration (R) _____ (True / False)		2
16.	Scarcity of water leads to a situation is called _____		2
17.	The process of extraction of ores from the mines is called _____		2
18.	Examples of non- renewable energy resources are _____, _____ & _____.		2
19.	Quinine is obtained from the bark of _____tree.		3
20.	The variety and variability among all living organisms is called _____		3

A - Answer **CO** – Course Outcome

MAHATMAGANDHIINSTITUTEOFTECHNOLOGY(Autonomous)
Kokapet(Village),Gandipet(Mandal),Hyderabad500075.

B.Tech.–SEMESTER-II **MR-22**

MID-TERM EXAMINATION-I

PART B : DESCRIPTIVE EXAMINATION

Course : **Environmental Science** Time : **90 Min**

Date & Session : **20/04/2024, AN** Maximum Marks : **20**

Branch & Section: **CIVIL, ECE, EEE, MECH, MCT, MME** Roll No:

Note: Answe rany FOUR full questions. All questions carry equal marks. **(4X5M = 20Marks)**

Q.No.	Stem of the Question	M	L	CO
1. a)	Explain the biotic and abiotic components of an ecosystem.	3	2	1
b)	Give an account of upstream and downstream problems of big dams.	2	3	2
2. a)	Define a biogeochemical cycle. Explain nitrogen cycle with a neat diagram.	3	1	1
b)	How is solar energy harnessed? Enumerate its applications.	2	3	2
3. a)	Discuss energy flow in an ecosystem.	3	3	1
b)	What are the effects of overutilization of surface and ground water?	2	2	2
4. a)	What is a flood? Give its consequences.	3	2	2
b)	Differentiate biomagnification and bioaccumulation.	2	3	1
5. a)	Explain genetic, species and ecosystem diversity.	3	3	3
b)	What are ecological pyramids? Explain why some are upright and some are inverted with suitable examples.	2	2	1
6. a)	Describe the various values of biodiversity.	3	2	3
b)	Discuss the impacts of mining activities on the environment.	2	1	2

M:Marks; L:Bloom'sTaxonomyLevel; CO:CourseOutcome

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)
(Village), Gandipet (Mandal), Hyderabad -500 075.

MR-22

ACADEMIC YEAR : 2023-24
B.Tech. – SEMESTER: II
MID - TERM EXAMINATION: I
PART A: OBJECTIVE EXAMINATION

Course : **Environmental Science** Time : **30 Min**
Date & Session : 20/04/2024, AN Maximum Marks :10
Branch & Section: ECE 1,2,3; CIVIL, EEE, MECH, MCT, MME Roll No:

Note: Answer All Questions. All Questions Carry Equal Marks.

Q.No.	Stem of the Question	A	CO
1.	The Progressive accumulation of non-biodegradable chemicals through food chain is known as (a) Ecological Balance (b) Biomagnification (c) Trophic Structure (d) Carrying Capacity		1
2.	Energy Flow in an Ecosystem is (a) Cyclic (b) Unidirectional (c) varies with the type of ecosystem (d) Two way		1
3.	The Process of conversion of nitrates to NH_3 during nitrogen fixation is called (a) Nitrification (b) Denitrification (c) Ammonification (d) Denitrification		1
4.	Pyramid of numbers in grassland ecosystem is (a) Diagonal (b) Inverted (c) Upright (d) Spherical		1
5.	Tidal Energy can be harnessed by construction of (a) Tidal Barrage (b) Tidal Column (c) Channels (d) None of the above		2
6.	Which of the following is an effect of overutilization of ground water (a) Decrease of Ground Water (b) Ground Subsidence (c) Earthquakes and Landslides (d) All of the above		2
7.	Identify the ecological use of Forests. (a) Reducing Global warming (b) Production of Oxygen (b) Both a & b are correct (d) Providing Commercial Products		2
8.	The process of extraction of minerals from the earth layers is called (a) Afforestation (b) Mining (c) Bioremediation (d) Biotic Potential		2
9.	The variety and variability among the groups of living organisms is called (a) Bioseperation (b) Bioaccumulation (c) Biodiversity (d) None of the above		3
10.	Identify the level of biodiversity among the following options (a) Genetic biodiversity (b) Species Biodiversity (c) Ecosystem biodiversity (d) All of the above		3
11.	_____ Food chain starts with dead organic matter.		1
12.	A network of food chains which are interconnected in an ecosystem is called _____		1
13.	_____ is the maximum population size of biological species that the environment can sustain.		1
14.	Biomagnification of pesticide _____ in the food chain resulted in thinning of egg shells.		1
15.	_____ & _____ are examples of renewable energy resources..		2
16.	The major constituent present in Natural gas is _____.		2
17.	Submergence of low lying area and human settlement is due to _____.		2
18.	One of the environmental damage due to mining is _____.		2
19.	"All Life must be Preserved" refers to the _____ value of biodiversity.		3
20.	Holy plants and Holy animals refers to the _____ value of Biodiversity		3

A - Answer **CO** – Course Outcome

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

Kokapet (Village), Gandipet (Mandal), Hyderabad -500 075.

MR-22

10

Academic Year: 2024-25

B.Tech. - SEMESTER - I

MID - TERM EXAMINATION - I

PART A: OBJECTIVE EXAMINATION

Course : **Environmental Science** Time : **30 Min**

Date & Session : 16/11/2024, FN Maximum Marks : **10**

Branch & Section: CSE, CSB, IT, CSM and CSD Roll No:

Note: Answer All Questions. All Questions Carry Equal Marks.

Q.No.	Stem of the Question	A	CO
1	Which of the following is a characteristic of ecosystems? a)Presence of only living organisms c)Energy flow through the ecosystem d) Exclusively composed of abiotic factors b)Static and unchanging environment		1
2	The most common usable nitrogen source is a) Nitrate b) Nitrite c) Nitrogen oxide d) Hydrogen		1
3	What is the role of oceans in the carbon cycle? a) Absorbs oxygen c)Absorbs CO_2 b) Release oxygen d) Release CO_2.		1
4	The role of producers in an ecosystem is to: a)Decompose organic matter c) Convert sunlight into chemical energy b)Consume other organisms for energy d) Break down inorganic nutrients		1
5	Large dams are not good for the environment as a) They create static water body that replaces a free-flowing river b) Inundation of valleys that submerge agricultural land, forests, habitations c) Accelerated sedimentation due to altered land use pattern d) All of the above		2
6	Tidal energy is a)Natural rise in sunlight c)Natural rise in Moon energy b)Natural rise and fall of tides d)Natural rise in water temperature		2
7	Aquifers are an important source of a)Rain b)River c)Fresh water d) Salt water		2
8	Which is not a correct statement a)Forests are habitat to all wild animals, plants and support millions of species b)Forest can act as pollution purifier by absorbing toxic gases c)Forest help in soil erosion d)Forest help in reducing global warming caused by greenhouse gases		2

(PTO)

9	Which of the following is NOT a component of biodiversity? a) Genetic diversity c) Climatic diversity b) Species diversity d) Ecosystem diversity		3
10	River Ganga is an example of (a) Consumptive value of biodiversity (b) Ethical value of biodiversity (c) Productive value of biodiversity (d) Aesthetic value of biodiversity		3
11	Conversion of nitrates to nitrogen is called as _____		1
12	Pyramid of energy is always upright T/F		1
13	The cycling of nutrients in an ecosystem _____ and _____ Components		1
14	Carnivores are the organisms occupy the highest trophic level in a food web T/F		1
15	_____ Elements are used in nuclear power.		2
16	_____ are devices is used to generate sunlight to electricity		2
17	The process of extraction and purification of ore from earth is called _____.		2
18	Methane (CH_4) is the main component of the natural gas. T/F		2
19	_____ is the drug prepared from Cinchona bark		3
20	A community of interacting groups from different species living in a single habitat is known as Ecosystem diversity T/F		3

A - Answer **CO** – Course Outcome

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)
Kokapet (Village), Gandipet (Mandal), Hyderabad- 500 075.

MR-22

Academic Year: 2024-25
B.Tech. – SEMESTER: I
MID-TERM EXAMINATION: I
PART B: DESCRIPTIVE EXAMINATION

Course : Environmental Science
Date & Session : 16/11/2024, FN
Branch & Section: CSE, CSB, IT, CSM and CSD

Time : **90 Min**
Maximum Marks : **20**
Roll No:

Note: Answer any FOUR full questions. All questions carry equal marks. **(4X5M=20 Marks)**

Q.No.	Stem of the Question	M	L	CO
1.a)	How to distinguish grazing from detritus food chain? Explain with suitable examples	3	3	1
b)	Enumerate the upstream and downstream problems associated with construction of large dams	2	2	2
2.a)	How Nitrogen fixation is an important functioning for proper ecosystem?	3	3	1
b)	Summarize the impacts of over mining on ecological and health. How to address this issue for protection of mineral resources	2	2	2
3.a)	What are renewable resources? How solar energy can fulfill the future needs and give their advantages	3	1	2
b)	Define primary production and secondary production. Discuss the single channel energy flow model of ecosystem	2	2	1
4.a)	Give classification of floods and droughts. Explain the various problems of droughts	3	2	2
b)	What are ecological pyramids? Describe the pond and grassland ecosystem of pyramid of biomass with suitable examples	2	1	1
5.a)	Explain the terms with suitable examples a) Genetic biodiversity b) Species biodiversity c) Ecosystem biodiversity	3	1	3
b)	Define and explain the following a) Biomagnification b) Carrying capacity	2	2	1
6.a)	What are social and ethical use values of biodiversity? How these values are conserving the ecosystem biodiversity	3	3	3
b)	Distinguish functions and benefits of forests .Summarize the strategies can be implemented to control deforestation	2	4	2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome